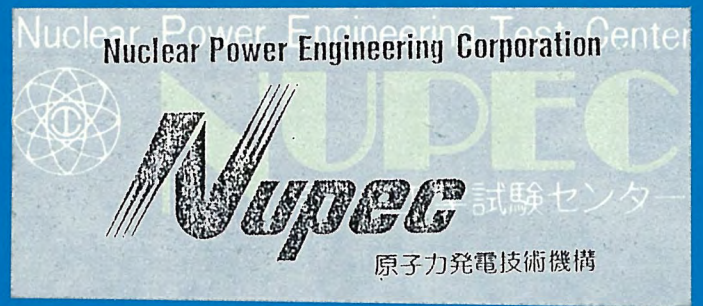




TAKASAGO

ENGINEERING LABORATORY



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OUTLINE OF THE TAKASAGO ENGINEERING LABORATORY

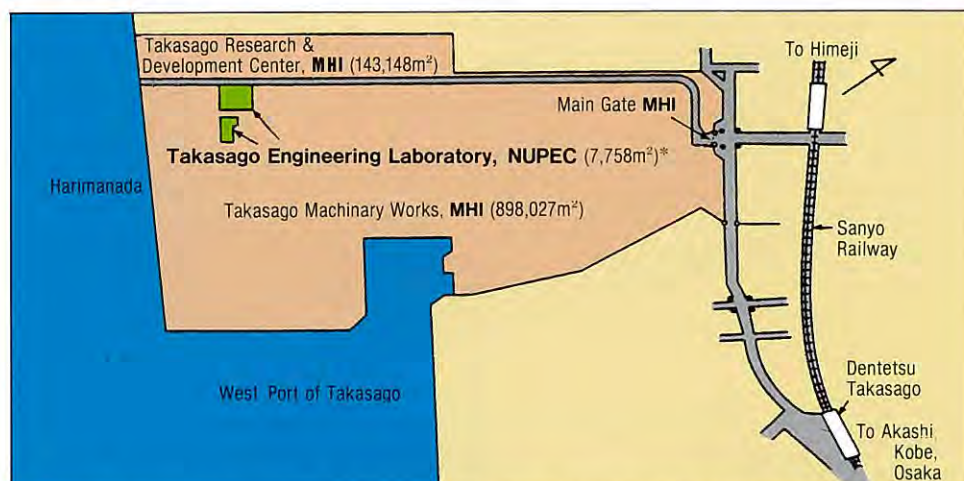
The Nuclear Power Engineering Test Center (**NUPEC**) has been performing the proving tests and verification tests on the safety and reliability of components and equipments for nuclear power plants and other activities under the sponsorship of the Ministry of International Trade and Industry (**MITI**) from March, 1976.

The Takasago Engineering Laboratory was established on November 1, 1983, for performing above mentioned tests of some **PWR** components.

■ Location and Site

The Takasago Engineering Laboratory is located at the south-west corner of the Takasago Machinery Works, Mitsubishi Heavy Industries, LTD. (**MHI**) and the ground area is 7,758m².

The location and site of this laboratory are shown as follows.



* 2-1-1 Shinhamada, Arai-cho, Takasago City, Hyogo Pref., 676 Japan. phone (0794)43-3663

PROVING TEST ON THE RELIABILITY FOR NUCLEAR FUEL ASSEMBLIES

VOID FRACTION MEASUREMENT TEST IN PWR FUEL ASSEMBLY

■ Purpose

The condition of the coolant, which flows through the fuel assembly, is very important in the operation of nuclear power reactors.

The behaviors of void (steam bubbles), which are occurring in the boiling condition of the coolant, largely affect the nuclear and thermal characteristics of the reactor, and are one of the most important factor in the core-fuel design.

In **PWR** system, there is little void during normal operation, but there would be remarkable void in a high power fuel assemblies at the critical operation state like abnormal transients and accidents.

This test is intending to prove the reliability of the void correlations, fill up the void of data base and improve the reliability of the analysis method by making clear the detailed void behaviors in rod bundles composed of plural fuel rods in the fuel assemblies used in actual reactors.

■ Test Procedure

This test is intending to measure the void fraction (ratio of steam to water) distribution in the fuel assembly by γ -ray using the thermal hydraulic test facilities that can simulate the same conditions as in the actual nuclear reactors.

At first, the test is made with an electrically heated Single Rod in order to fill up the basic void data.

Next, the test is made on electrically heated 5×5 Test Bundles that can simulate the same conditions as 17×17 type **PWR** Fuel Assembly used in actual reactors.

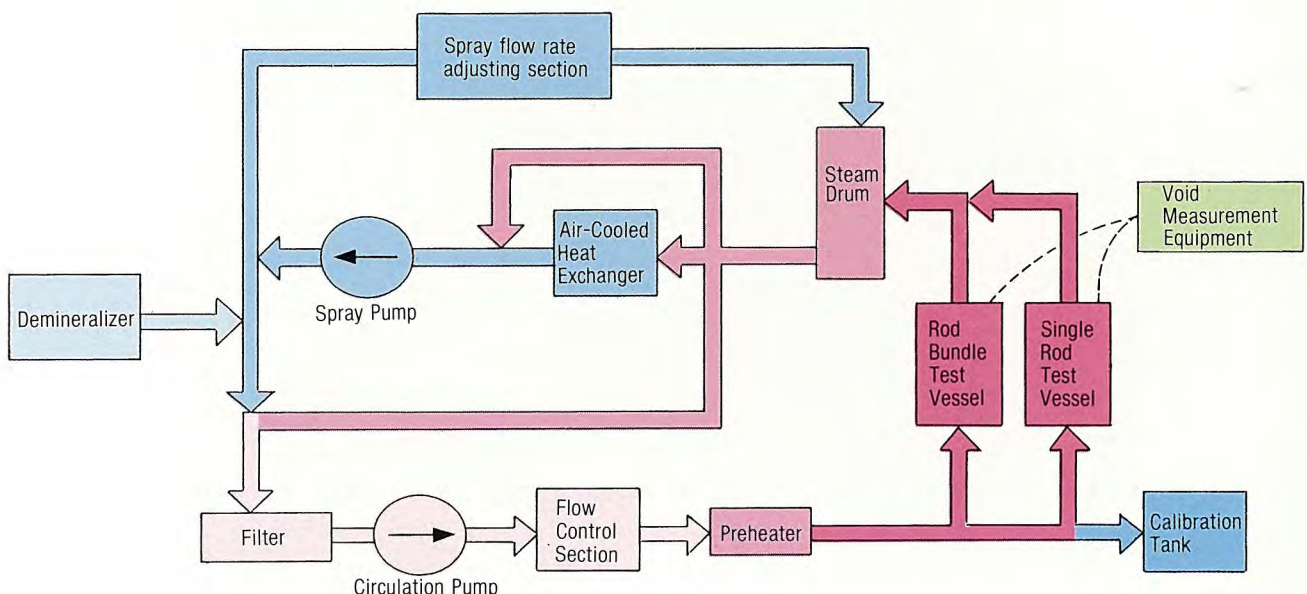
■ Test Facilities

The **DNB** Test Facilities are partially remodelled to use as Void Measurement Test Facilities.

The main remodelled sections are the Single Rod Test Vessel, the Rod Bundle Test Vessel and the Void Measuring Equipment.

Test facilities are shown in the following diagram.

● System diagram of **PWR** Fuel Assembly Void Measurement Test Facilities.



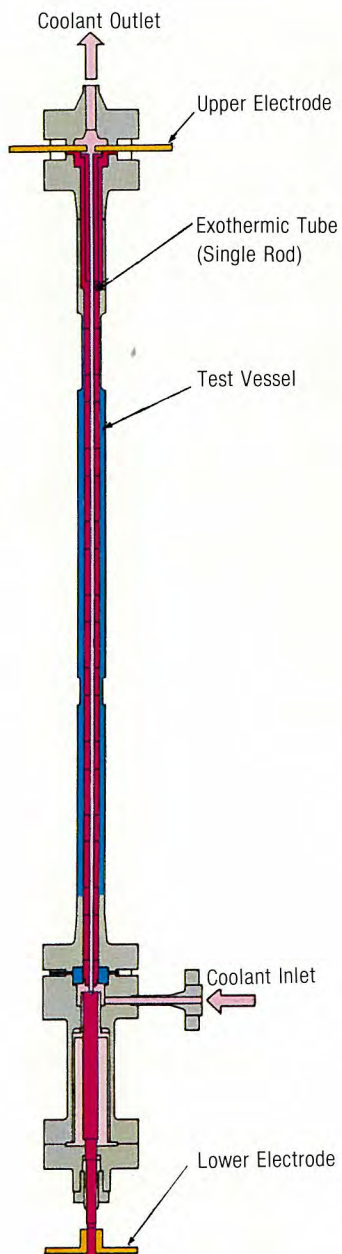
- Max. Heating power : • Single Rod Test 150Kw
- Rod Bundle Test 10Mw (Approx. 500% Rated output of fuel assembly.)
- Max. Coolant flow rate : 60t/h (Approx. 150% Rated coolant flow through fuel assembly.)
- Max. Coolant pressure : 195Kg/cm²

● Single Rod Test Vessel and Void Measurement Equipment

In this test, void fraction measures in single cell is simulating the same coolant conditions that in the fuel assembly by inserting an Exothermic Tube (Single Rod) into the Test Vessel.

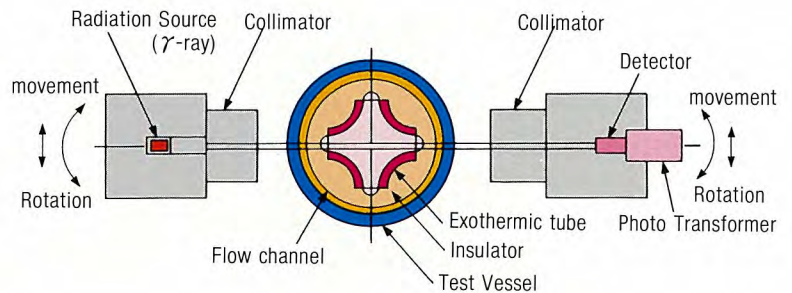
The Test Vessel is made of Titanium Alloy to minimize the γ -ray attenuation in the structure.

● Single Rod Test Vessel

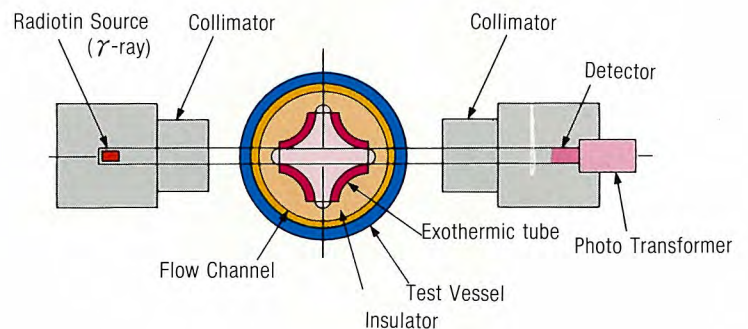


● Void Measurement Method

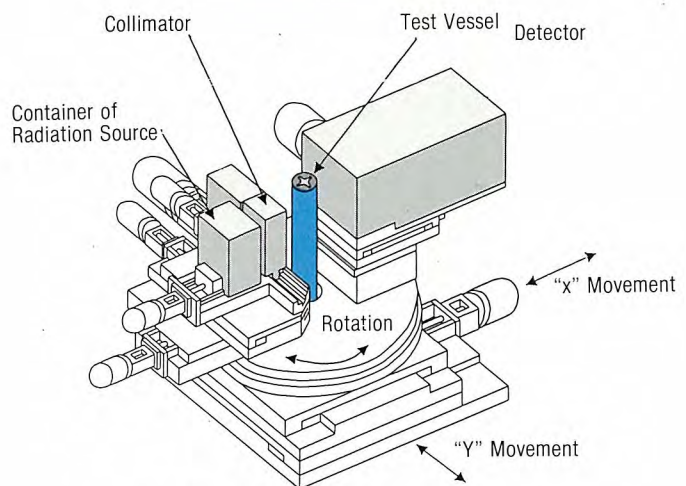
(1) Measurement by Narrow Beam.



(2) Measurement by Wide Beam.



● Void Measurement Equipment



Note : This Equipment shows "X" axis only.

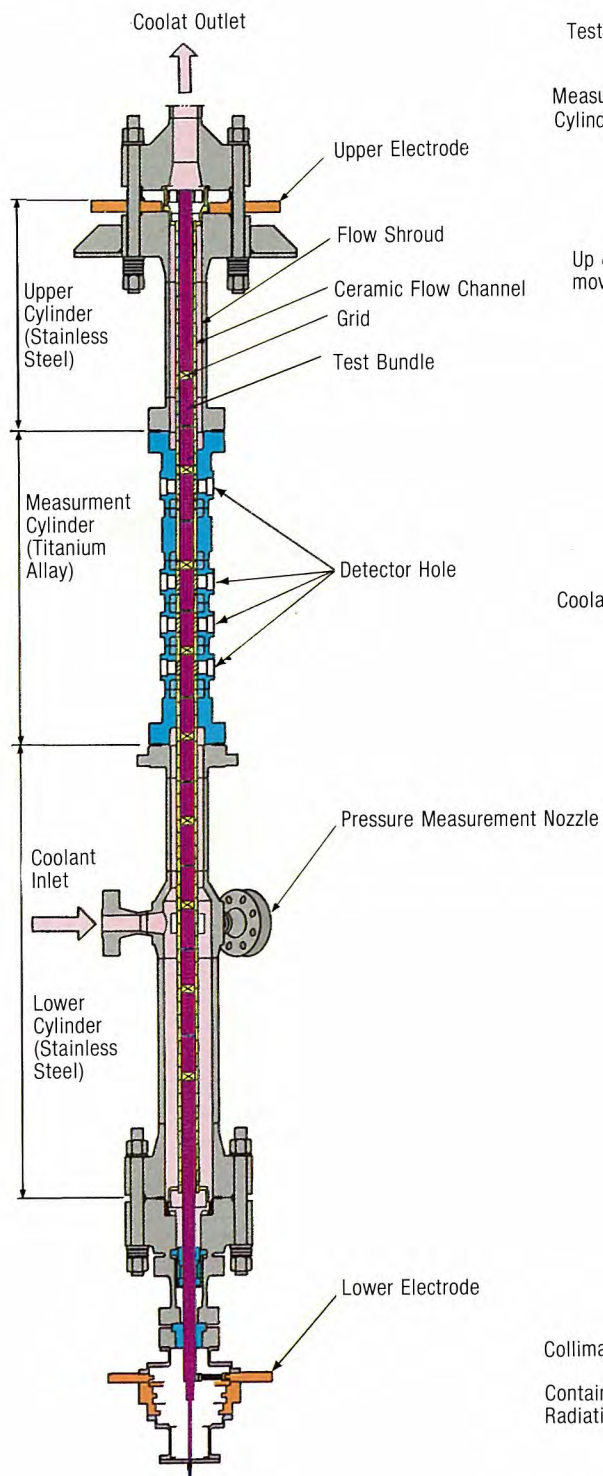
● Rod Bundle Test Vessel and Void Measurement Equipment

This Test Vessel consists of the Upper Cylinder, the Measurement Cylinder and the Lower Cylinder.

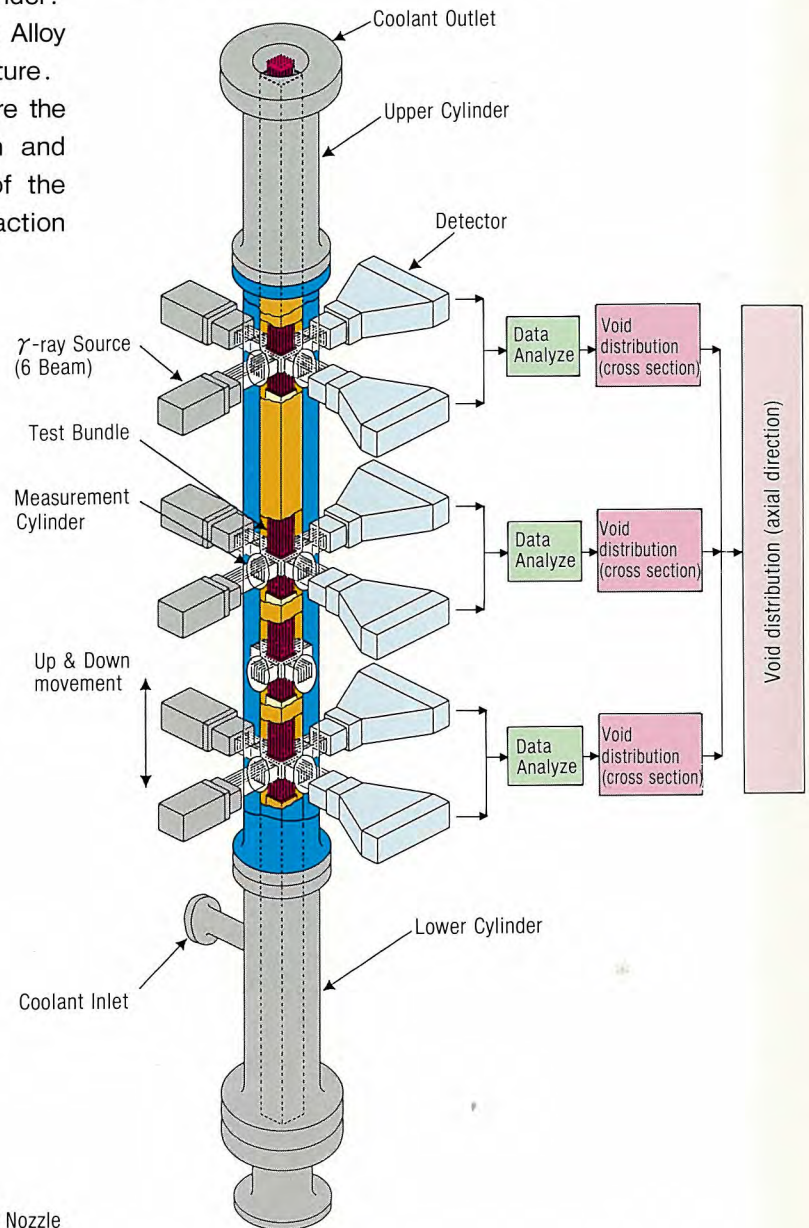
The Measurement Cylinder is made of Titanium Alloy to minimize the γ -ray attenuation in the structure.

The Void Measurement Equipment can measure the void fraction distribution of the cross section and in the axial direction in the three positions of the Test Bundle, and can analyze the void fraction distribution in three dimensions.

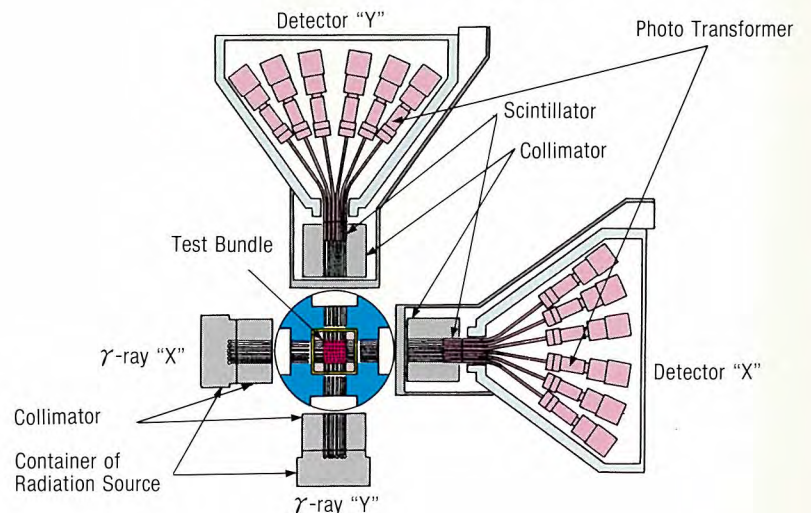
● Test Vessel



● Void Measurement Equipment



● Section of Measurement Equipment



■ Test Bundles

Test Bundles are fabricated by simulating the partial section and full length of the actual 17×17 type PWR Fuel Assembly.

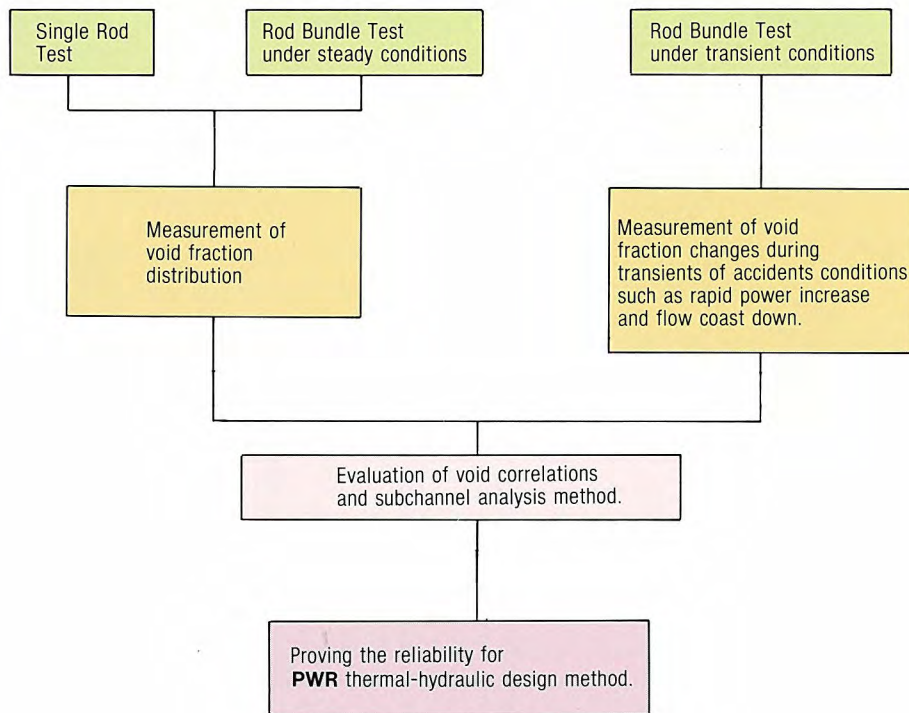
These are arranged in 5×5 square array having the following specifications and type.

● Specifications of Test Bundles

Spec. Test Bundle No.	Heater Rod array arrangement	Outer dia of Heater Rod (mm)	Outer dia of Guide Thimble (mm)	Type of Cell	Axial power distribution	Effective length of Heater Rod (mm)
1	5×5	9.50	—	Typical	Uniform	3,660
2	5×5	9.50	—	Typical	Cosine	3,660
3	5×5	9.50	12.24	Thimble	Cosine	3,660

Note; Thermocouples are attached to inner surface of Heater Rods to detect the boiling transition. Test Bundle is installed within the Test Vessel, and Heater Rods are heated by upper and lower electrodes applying direct current under simulated actual operating flow conditions.

■ Test Items and Evaluation



■ Program Schedule

Fiscal Year	1987	1988	1989	1990	1991	1992	1993
Design							
Construction of Test Facilities							
Fabrication of Test Bundles							
Test and Evaluation							

PROVING TEST ON THE RELIABILITY FOR NUCLEAR FUEL ASSEMBLIES

PWR FUEL ASSEMBLY MAXIMUM THERMAL LOADING TEST (DNB TEST)

■ Purpose

In PWR system, primary coolant is heated by fuel assemblies in the reactor core, and the amount of heat which is transferred from the fuel rod to the coolant depends on the flow rate, the temperature and the pressure of the coolant and on the surface temperature of the fuel rod.

It is necessary to maintain the heat flux below the **DNB** (Departure from Nucleate Boiling) Heat Flux for safety reason because the burn out of the fuel rod occurs when the heat flux exceed the **DNB** Heat Flux.

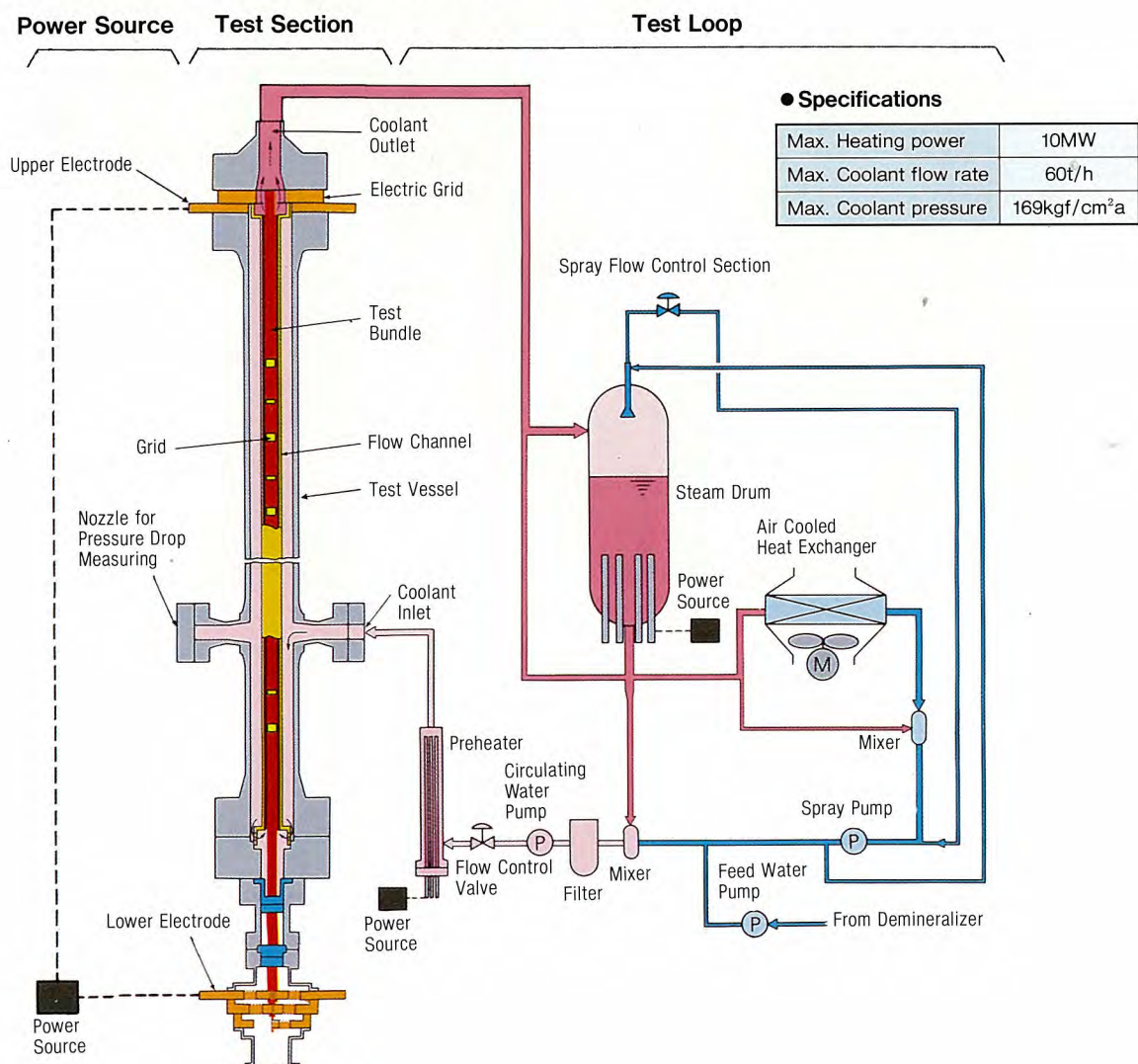
Therefore, the fuel rod must be maintained, its integrity during reactor operation having enough margin against the **DNB** Heat Flux keeping Forced convection and/or Nucleate boiling heat transfer condition.

This proving test was intended to prove the reliability of fuel assembly design by confirming the thermal margin of heat transfer.

■ Test Facilities

These Test Facilities were designed to simulate the thermal flow conditions in the actual nuclear power reactors and include mainly two systems: a Circulating system to supply hot pressurized water for the tests, and a Cooling system to remove the heat generated in the tests and to control the test temperature by using a Heat Exchanger. Auxiliary facilities include feedwater supply and flow calibration systems.

● System diagram of the DNB Test Facilities



■ Test Bundles

Test Bundles were fabricated by simulating the partial section and full length of the actual 17X17 type **PWR** Fuel Assembly. These are arranged in 5X5 and 6X6 square array having the following specifications and type.

● Specifications of Test Bundles

Items	Heater Rod array arrangement	Outer dia of Heater Rod (mm)	Outer dia of Guide Thimble (mm)	Pitch of Heater Rod center (mm)	Effective length of Heater Rod (mm)	Number of Heater Rods	
						Typical	Thimble
Spec.	5X5 (6X6)	9.50	12.24	12.60	3,660	25 (36)	24 (35)

Figures in the parenthesis show the number of Heater Rods in the case of 6X6 Heater Rod array arrangement.

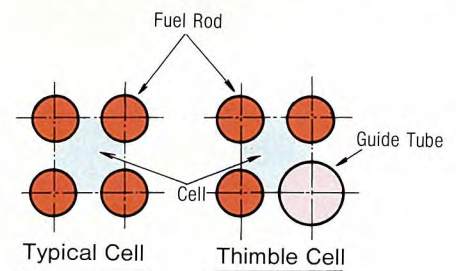
● Type of Test Bundles

Test Bundle No.	1	2	3	4	5	6	7	8	9	10	11	12
Items												
Rod array arrangement	5X5	5X5	6X6	5X5	5X5	6X6	5X5	5X5	5X5	5X5	5X5	5X5
Number of Grids	9	9	9	9	9	9	9	9	9	9	9	9
Type of Cell	Typical	Typical	Typical	Typical	Typical	Thimble	Thimble	Thimble	Thimble	Thimble	Typical	Thimble
Axial power distribution	Uniform	Uniform	Uniform	Cosine	Cosine	Uniform	Cosine	Cosine	Cosine	Cosine	Cosine	Cosine
Type of Test	Mixing Test	Steady State Test							Test with Bowed Rod.		Transient Test	

● Definition

Typical Cell: Cell surrounded by four Fuel Rods

Thimble Cell: Cell surrounded by three Fuel Rods and one Guide Tube for Control Rod or In-Core Monitor



■ Test Procedure and Evaluation

Test Bundle was installed within the Test Section, and Heater Rods were heated by upper and lower electrodes applying direct current under simulated actual operating flow conditions.

Various data such as coolant temperature, pressure, pressure drop through Test Bundle and other pertinent data were measured.

Thermocouples were attached to inner surface of Heater Rods to detect the boiling transition.

Test Items and Evaluation procedure are shown as follows.

● Test Items

Items	Evaluation
DNB Test in Typical Cell	Thermal margin at normal and transient operation were evaluated by DNB Test under actual operating conditions using Test Bundles simulated actual Typical Cell of Fuel Assembly.
DNB Test in Thimble Cell	Thermal margin at normal and transient operation were evaluated by DNB Test under actual operating conditions using Test Bundles simulated actual Thimble Cell of Fuel Assembly.
DNB Test in Cell including Bowed Rod	Verified the effect of Bowed Rod for the DNB Heat Flux using Test Bundles simulated actual Fuel Assembly including Bowed Fuel Rod.
Coolant mixing Test in Typical Cell	Measured the mixing ratio of coolant under actual operating conditions using Test Bundles simulated actual Fuel Assembly.

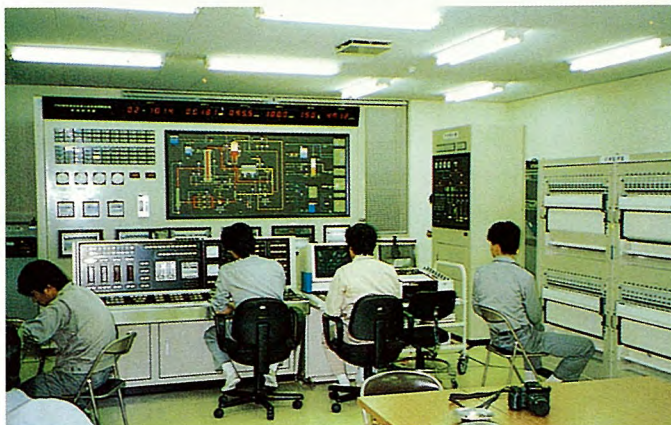
■ Test Scene



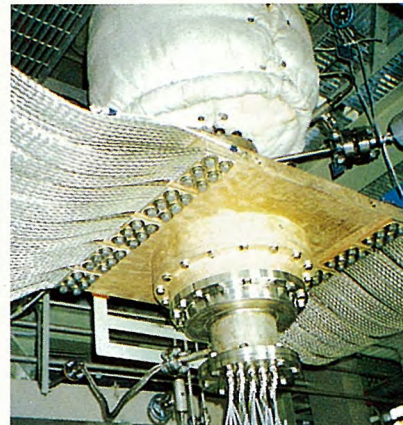
Test Bundle under loading



Upper Electrode



Control room in operation



Lower Electrode

■ Test Results

An extensive experimental program was performed to investigate the behavior of **DNB** in electrically heated Rod Bundles simulating a portion of **PWR** Fuel Assembly.

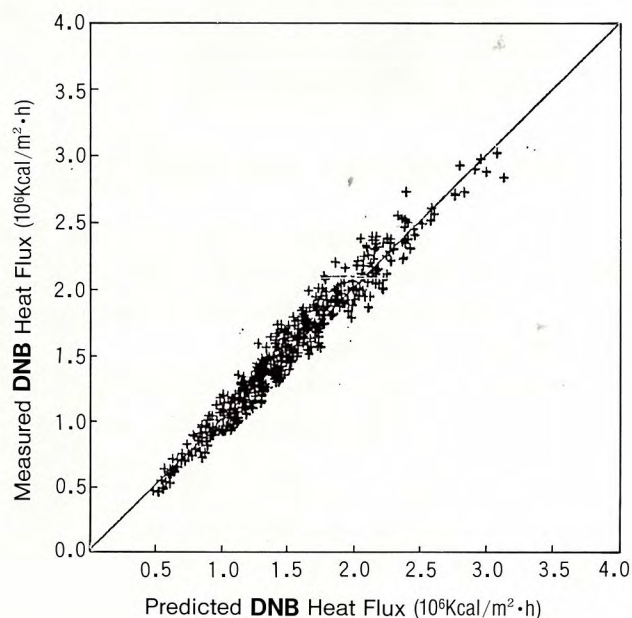
About 900 **DNB** data were analyzed.

The results were presented as Measured **DNB** Heat Flux versus Predicted **DNB** Heat Flux as shown in the right figure.

Good agreement was obtained for all of the test data normal and transient conditions.

It has been proved that **DNB** Heat Flux can be satisfactorily predicted by the current **DNB** Correlation of **PWR** Fuel Assembly.

Evaluating all of these test results, we found that the current thermal hydraulic design method used for **PWR** Fuel Assembly have sufficient reliability.



■ Program Schedule

Fiscal Year	1978	1979	1980	1981	1982	1983	1984	1985	1986	1987	1988	1989
Items												
Design												
Construction of Test Facilities and fabrication of Test Bundles												
Test and Evaluation												